

FACULTY MEETING 11/5/2021

Vikas, Alan, Meenesh, Anh, Betul, Sangil, Shafigh, Lew, Michael, Said

Expenditures corrected and updated

Citations and papers published trending up from previous years. **Add publications to Google Scholar**

Research funding

BS & MS in Molecular Engineering

- curriculum development-resource assessment

- Setting a committee to build a BS & MS (courses or thesis) program. Need to clean up the frame for the program. Clinical faculty required, courses...

NSF funded IUSE proposal on Institutional Transformation via Department Action Teams (DAT) To begin in Spring 2023 – curriculum improvement – other transformational activities – funds will be available – looking for ideas

CHE Entrepreneurs club → Engagement with DPI (DPI is funding a lot of computational research)

- Provost is doing data collection on entrepreneurial activities and ours is the only one in the COE

Climate survey for faculty will be distributed soon. Diversity strategic plan for the plan. Collecting info from students and faculty

Alan

Advising week. iAdvise feedback: Not everyone signs up, even after reminders. What can be the process to deal with the latecomers?

- 70% completion during advising week

- Use harsher language not in actions.

High school outreach, 1 wants to visit EIB and 3 others want a school visit. Market ourselves directly to schools do not rely on broad

CHEME Industry Day November 12th Panel of professionals at different

Adjunct Profs. Kyle Kostroski Nov. 9th at 11 am & Jake Edie (494 CHE) 11/16

Naveen Dandu (research assistant professor) will be teaching 454

Adjuncts for summer courses needed

ChemE 150 approved under “Understanding the natural world” and will move onto the next stage

CHE312: Sangil Kim

Receiving applications already and working on 50 odd list of students interested in Grad school

3 + 2 program has gone down but is still active

Diversity Committee: Advancing Racial Equity

Student(1 grad student included) committee created and putting up action items and ideas

ChemE Culture club is highlighting the diversity of the department

Climate survey

Initiative finding wanted at the end of December and then going to college then university

Concerns about adjuncts teaching core courses _primary reason is due to scarcity of faculty

Working on requesting more positions from the college and expect to see it soon

Midwest Regional Conference AIChE is at UIC and

Executive Committee/ Lew interviews with new Associate Deans Carmen Lilley (grad students) Karl Rockney (research development)

MHub & iLab

Provost is very excited about entrepreneurial endeavors and if CHE could take a leadership position it would be favorable

COMPLETE CLIMATE SURVEY

https://uic.ca1.qualtrics.com/jfe/form/SV_a36UQQAy5Srno

December 11th Faculty Meeting

Vikas Berry, [Said Al-Hallaj](#), [Michael Caracotsios](#), Lew Wedgewood, Brian Chaplin, Anh Ngo, Gang Cheng, Ying Liu, Alan Zdunek, Sangil Kim, Betul Bilgin, Shafigh Mehraeen, Christos Takoudis

Minutes passed with Alan Zdunek edits

CREDIT/NONCREDIT

- Discussion of credit & non-credit, a bit unfair for students whose GPA will be affected.
- There can be NO retroactive action, don't change grade on request. Cannot lower their grade to benefit their gpa
- Pete will create a statement regarding policy

Ezinne Achinivu-Faculty Position

- Opportunity hire. Need to diversify in terms of research, culture and gender
- Very clear presentation, shows good aptitude for teaching
- Want to make sure the administration is on the same page as us in hiring someone to achieve tenure
- Technical things not adding up to research/CV and technical questions avoided
- Publications are not there
- Are we making concessions for cultural/gender diversity? Will we then make these concessions when tenure is sought?
- We don't want to lower the bar for people eventually seeking tenure?
- Moving from energy to materials due to funding
- WISEST always makes an effort to have a diverse pool of candidates but the end goal is to hire on Merit. Only when you have equal candidates should you lean to a diverse candidate hire
- Mentoring program should be developed to aid new hires
- Can she be mentored to become successful
- Concerns about research direction and development
- Where is her point of competing with independent laboratories or industry? Where is she able to collaborate? She may need help in developing collaboration and support
- What is the likeliness of us getting a similar candidate as them
- She has a solid understanding and reasoning for questions posed to research
- She's making connections. Holistic vision
- This is not an open search. This hire will come from the chancellor
- No high impact publications, she needs someone to help give her direction
- She still feels very new postdoc
- Anh has 2 diverse candidates that will be sent for review
- She's going to be a good teacher she has the knowledge and vision for it
- Short term research is great but long term is in the air
- **Position for Opportunity Hire:**
 - There is a mentorship program by the college
 - We don't know how long the position will be open

- We have to ask the chancellor for funding before we can make the offer (president's money)
 - Deadline? No because it's not an advertised position
- If we have a mentorship program in mind, would we consider hiring PhD candidates? Yes, but it's hard to attract the ones on our seminar series. It's hard to compete as many of them are "stars"
- Vote on pursuing Hire
 - Aye: IIII
 - Naye: I

Clinical Track Faculty Search

- Alan will chair
- Waiting on college HR approval to move forward

Department Name

- No name change
- # 1 Department of Chemical Engineering & Nanotechnology
- 10 options
- How would it affect our recruitment (undergrad especially), impression on peer departments, student impression
- Department name changes but not their degree
- Nanotechnology is a bit dated. Perhaps Quantum technology?
- Chemical engineering is not clear to undergrads or high school students. What does it mean?
- Nanotechnology is a buzz word for recruitment
- Competing with other departments or universities so we might as well not change
- Marketing & descriptive standpoint
- Perhaps "chemical & molecular engineering"
- Helps our academics but not our recruitment or comprehension
- Perhaps invest in marketing our new tracks and describing what chemical engineering is.
- Civil engineering changed to Civil & Environmental Engineering and they stole some of the students we usually got
- Adding to the name will not hurt
- "Electrochemical" added in
- Chemical Engineering is more easily recognized despite it being initially known as gas and oil

CHE 100

- **CHEME Industry Day**
 - 2/26, 3/5 3/12 possible dates to see if people are having exams
 - Encourage access students

Undergraduate Student Office

- Climate warming course approved

- 301 concurrent curriculum change approved
- 230 add approved
- Said has approved 494 Entrepreneurship as a permanent course:
 - College had a separate course and they filled the gap between the groups
 - 20 students and the quality was incredible
 - Next step to work with faculty to commercialize research developments and make it a senior project (needs a lot of time)
 - Only a fall course
 - Basics accounting, ROI
 - Motion to approve permanency of course
 - The Ayes carry it
- Meenesh has 494 biopharmaceutical... this will also be part of the certificate program
- 494 Clean Energy and the grid from Said in the Spring Interface of energy storage and the smart grid
- Change name of 433
- ABET course surveys and codes have been sent out and faculty should have been received
- 330 course change approval in exchange for p-chem. Modification of major please sign and send to Alan
- Discuss the necessity of the ABET survey to the department

Graduate Student Office

- Started to receive applications for next year. Mainly MS program
- Just 3 phd applications
- 3 students abroad attempting to get visas, tuition will be waived if they still can't come in the spring and TA position offers will be honored when they do come.

FACULTY MEETING 2/4/2022

Vikas, Said, Alan, Anh, Betul, Michael, Shafigh, Brian, Sangil, Ying, Lew, Meenesh

- Minutes 12.03.21 Passed
- Naveen Dandu new Research Assistant Professor
- BS & MS in Molecular Engineering (Vivek Committee Chair) what courses and resources are needed?
 - We would eventually need to get accreditation
- Need to hire more adjunct and 1 more clinical track
 - Jake Edie teaching climate course this semester
 - Kyle Kostroski CHE 313 in the fall(BP)
 - 1/9th salary for teaching summer course
- CHE 150 Climate Change course approved and listed for the fall semester
- Pete will be visiting EIB to see space constraints
- Research Symposium will be hosted this semester
 - Remote? In-person?
 - Remote exhaustion
 - Academic/Professional Panel, OVCR presentation, Research presentations, poster board presentations
- Social Media Videos from department faculty and students to showcase lab work, student life, and research
- Discussion of Students' PhD Thesis
 - If student publishes a paper it can be added into a thesis
 - Coauthoring must be clarified what was done by each
 - Section on lab contributions
 - Careful of overlap in thesis within lab
- Enrollments
 - Enrollment trends in all CHE departments across USA has dropped , possibly due to CS rise
 - 23.6% down in the past 3 years 4+1 program has helped to mitigate this issue
 - 17% down in application this year
 - Social media strategy for marketing department : Post Alumni success stories, senior design project videos
 - Profession has a bad image. Promote EV, Environmental, Food, molecular, nano, biotech
 - Bring along successful Alumni to recruitment
 - Have Alumni seminars in freshman/sophomore year
 - Impressions from the students is that we have
 - Purdue has great undergrad research experience
 - **Committee for Retention and Recruitment meeting**
- CHE Student Climate Survey
 - 76 undergraduate & 8 graduate
 - 40- 60 female-male

- 24% Latino, 33% white, 22% Asian, 7% arab, 9% other
- Feel well included in the department
- 32% feel they don't have enough Social/emotional support & about 10% need academic support
- 13% have experienced or witnessed racial microaggression
- Vast majority say they were comfortable with the department climate
- Create more inclusive clubs
- More student-faculty interactions and engagements
- FACULTY CLIMATE SURVEY
- Lack of response, No female reported
- Feel well supported, Feel it is a respectful, unbiased, not racist, no harassment
- Opportunities at CHE for Professional success needs more access to professional development
- Student ARE committee
- Undergrad
 - Adjunct Prof suggestions are being accepted
 - Education Policy Committee: Engineering practicum level course for undergrads, encouraged by OIS to give international students internship opportunities
- Applied 27 MS/ 23 PhD
- Admitted 24/ 5 PhD
- Rejected 3/2
- Pending review 0/15
- Deadline for international is 2/15 and about a month later for domestic
- Entrepreneurship Club
 - Janai is first minor in Entrepreneurship and is employed at mHub
 - Pilot program has 2 current positions and will open with more opportunities. Then eventually start looking for positions in other companies or government entities
- Conflict of Interest paperwork (RNUA)
 - Submit paperwork when holding any position or holding ownership
 - Can own shares but cannot have a paid position
 - DPI
- Order process
 - Tax waiver issue for vendors
 - Vendors don't have updated contact info and this is usually not communicated to the university system.
 - Cutting out huge vendors
 - Ordering Memo will be created
- Having more access to remote education but undergrad courses will remain in person

Minutes of faculty meeting (March 11, 2022)

- Provide feedback on courses on google sheet
- Courses AY2023:
 - Overlap on the courses between BS and MS
- Planning on research symposium: Naveen
 - Faculty participation in the panel discussions
 - Select a topic from agenda
- Ideas to share on google doc, to share what others are doing in terms of outreach activities
 - Helps Ezinne for ideas and others
- Planning a curriculum day
 - What is covered in the courses?
 - Course topics that are covered
 - Make plans to make sure it is updated/marketable for students benefit
 - Sarai is scheduling (may end is deadline)
 - 3 h of time (discuss syllabus)
 - Another meeting to see any needed changes!
- Social media videos (15 sec); Instagram
 - Helps recruiting
- Research equipment programs 5k
 - Using funds to help as many as possible
 - Servers for the departments or any equipment?
 - Is 5k enough?
 - Ying Liu: educational purposes (good idea)
 - Ezinne mentioned Autoclaves
 - Any other ideas?
 - Extending the deadline is possible, it was today!
- Betul Bilgin planning a summer program
 - Possibility for collaboration
 - REU students
 - 8-10 STUDENTS
 - Proposal due in august,
 - Any project to include in this proposal?
 - Program start in summer 2023; 10 week program
 - Middle schoolers and high schoolers working in the lab
- CHE 101: intro to ChE for the fall 2023.
 - Mandatory; 1 day per week; 0 credit
 - In the fall semester, voted to include in the curriculum
 - Each faculty presents their research or anything imp for ChE
 - 1 faculty presentation per week
 - Improved 33% in enrollment for CME department
- CHE 401: (planning for next meeting)

- Google documents (planning for next meeting)
 - Don't change anything
 - Let Vikas make the changes
 - Vikas will send email with the link

Undergrad:

By Alan Zdunek:

1. Fall 2022 Applications
 - a. Fall 2022- CHE majors: 91 admitted, 8 accepted offer
 - b. Fall 2022- undeclared majors: 327 admitted, 8 accepted
 - c. IGNITE event will be in-person- April 2nd – Resource Fair with College tables at EIB parking area and back EIB area
2. BS CHE degree changes approved by SCEP (Sub-Committee for Education Policy)- UIC Senate next.
 - a. CHEM 232 and 234 decreased to 3 credit hours each
 - b. CHE 396 decreased to 3 credit hours
 - c. CHE 230 added to degree requirements
3. Adjunct Hiring & Summer Session:
 - a. Betul Bilgin to teach CHE 313 Summer Session
 - b. Need instructor to teach CHE 312 in summer session
4. Advising Week
 - a. In-person or remote advising?
 - b. Using iAdvise and uDegree Audit
 - c. FAQ's and course schedule/flowchart sent
 - d. Adviser evaluation survey
5. ABET information
 - a. Review Program Educational Objectives- send survey
 - b. Course assessments and improvements- this semester
 - c. External Advisory Board meeting

Graduate admissions:

By Gang:

- Fall 2022: 12 PhD students accepted offer!
- Eligibility for TA slots
- Gang will send out the list to faculty

Minutes of faculty meeting (April 22, 2022)

1. Tracking strike is not permitted
 - a. Can only ask students that what they are doing today
 - b. If TA/RA is not working, please have track of it
2. Equipment update:
 - a. Poster printer ordered for the department
 - b. 3D printer
 - i. 3L is about 16K \$ + more for accessories
 - ii. There is a small version ver3.0 (4500)
 1. Vikas has one in his lab, if interested we can buy same
 2. Share the 3D printer across
 - iii. Sangil suggested less expensive 3D printer
 - iv. Meenesh suggested that department can use their 3D printer
 - v. Donating old printer ver2
 - c. A coffee machine can be bought as well
3. UIC institutional proposals: sharing what is going on this area
 - a. Ying Liu can send whitepaper to review
 - b. Areas:
 - i. Materials science
 - ii. Water
 - iii. Biopharma
 - iv. Decarbonization
4. Budget FY 2022
 - a. About 250K\$ is in reserves
 - i. ICR (86K\$)
 - ii. Gift (37K\$) – lesser than previous years
 - iii. Unit OP (~9K\$)
 - iv. 3+2 program and other MS (~132K\$)
5. Academic analytics and ranking:
 - a. We are not doing well in some areas as we are relatively younger faculty
 - b. As per total articles per years per faculty per department, we are about 10-12
 - c. Average publications per year is low; we need to publish at a rate of 100-200 per year
 - d. Citations per publication wise we are doing not bad (3rd quarter); we can do better
 - e. Grants wise we are low; we need to work on it. (Analytics shown is however doesn't look right). Vikas will check and get back to us.
 - f. US Department ranking – 61
 - i. Score is 2.5
 - ii. We have improved over the last few years

- iii. We need to do better, can compete with NCSU, NC, Ohio St, U.Wash
seattle
 - iv. QS stats show that we are #34 in US.
 - 6. Summer teaching
 - a. CHE 312: Jeffrey Zlac
 - b. CHE 313: Betul
 - 7. Department events:
 - a. Great achievements
 - i. By Meneesh
 - ii. Graduate symposium
 - iii. We have in-person Expo this week

Gang Cheng: (Graduate students admission report)

- 17 PhD students were admitted, 12 are in waiting list
- For masters – 30 students

Alan Zdunek: (Undergraduate report)

- 6. Fall 2022 Applications (4/19)
 - a. Fall 2022- CHE majors: 111 admitted, 28 accepted offer
 - b. Fall 2022- undeclared majors: 446 admitted, 73 accepted
 - c. IGNITE event held April 2nd College of Engineering sessions at EIB
 - i. CHE unit ops lab tour, presentation, Good turnout
- 7. BS CHE degree changes
 - a. UIC Senate approved 4/21/22
 - b. CHE 396 credit hours variable: 3-4 credit hours to handle transition to new curriculum requirements
- 8. Adjunct Hiring & Summer Session:
 - c. Dr. Jeffrey Zalc to teach CHE 312 in summer session-
- 9. AIChE Chicago Section McCormack Award-UIC
 - a. Paul Diaz, Senior
 - b. Student poster competition
- 10. ABET information
 - a. Review Program Educational Objectives- sent survey
 - i. Singh, Sharma, Kim, Cheng, Liu missing
 - b. Course rubric assessments- this semester
 - i. Save one set of course output to assess (all students)
 - c. External Advisory Board meeting- Summer 2022

8. Vikas:
 - a. Everyone is required to come up with their syllabus examples of problems they give in exams and projects.
 - b. It is in mid-May or end of May, an email will be sent out soon
9. Brian:
 - a. 4+1 program
 - i. Students contacted Eddie that tuition is raised from undergrad to grad without warning (3K \$ more)
 - ii. Hinted to file lawsuit against dept
 - iii. Can hurt 4+1 program!
 - iv. Vikas said: department is working on it, negotiating with grad school
 - v. The website must clearly mention that tuition is changed
10. Students: about Diversity Equity and Inclusion members:
 - a. Daniel Christiansen, Abigail Hartzel, Daisaq Cyrilla and Mario Ocampo
 - b. Discussed with Vikas and Bitul
 - c. Get feedback from all the faculty for the DEI improvements
 - d. Working on climate survey, trainings, workshops
 - e. Want to know if faculty are facing any challenges
 - f. To understand your implicit bias, Vikas suggested to take this test via the below link. <https://implicit.harvard.edu/implicit/takeatest.html>